Revision Session – Supervised and Unsupervised Learning Algorithms

Use the following dataset representing 10 students' performance in a course. Features include:

- Study Hours (weekly average hours studied)
- Attendance Percentage (percentage of classes attended)
- Previous Exam Score (out of 100)
- Gender (Male or Female)
- Extracurricular (Yes or No, indicating participation in activities)

Student ID	Study Hours	Attendance Percentage	Previous Exam Score	Gender	Extra-Curricular Activities	Result
1	5	80	65	Male	No	Pass
2	3	60	50	Female	No	Fail
3	6	70	70	Female	Yes	Pass
4	1	50	40	Male	Yes	Fail
5	4	75	60	Male	Yes	Fail
6	8	90	85	Female	Yes	Fail
7	7	85	80	Male	No	Pass
8	9	95	90	Female	Yes	Pass
9	2	80	50	Male	Yes	Fail
10	10	98	95	Male	Yes	Pass

a) Identify which columns denote categorical and numerical features.

b) Calculate the Gini index for "Gender" and "Extra-curricular Activities" feature columns. Show the steps in your calculations.

c) Based on part (b) result, which feature is most suited to take as the root node in generating the decision tree? Justify your answer.

d) The above dataset was split into training and testing sets, with the first 5 rows used for training and the remaining rows for testing. Based on a classifier trained on this split, the predicted values for the test set (Students 6 to 10) are provided.

Student ID	Actual Result	Predicted Result
6	Fail	Pass
7	Pass	Fail
8	Pass	Pass
9	Fail	Fail
10	Pass	Fail

i. Create the confusion matrix.
ii. Calculate the Accuracy of the classifier.
iii. Interpret what each metric Accuracy, Precision, Recall and F1-Score indicate about the classifier's performance in predicting student passing.

e) Calculate the means and standard deviations for each numerical column you identified in part (a).

f) Create the scaled dataset using Standardization method only for the numerical columns. ($Z = \frac{x-\mu}{\sigma}$)

g) Compute the Euclidean distances between Student 1 and Student 6 and Student 1 and Student 9 using the standardized dataset. Show the steps in your calculations.

h) Based on the results from part (g), identify which student is the most suitable to merge with Student 1 into the same cluster. Justify your answer.

i) For k-means clustering (k=2) on the standardized dataset obtained in part(f), the clusters and their centroids are given below.

Cluster	Student ID	Centroid Coordinates
Cluster 1 – C1	1, 2, 3, 4, 5	(-0.871, -0.785, -0.867)
Cluster 2 – C2	6, 7, 8, 9, 10	(0.871, 0.785, 0.867)

Answer the following:

i. Compute the inertia (within-cluster sum of squares) for the two clusters separately using the equation given below. Show calculations for each cluster and sum them to get the total inertia.

$$\boldsymbol{Inertia} = \sum_{\boldsymbol{points\ in\ cluster}} (\boldsymbol{Euclidean\ distance\ to\ centroid})^2$$

ii. Compute the Dunn index for the above clustering. Show all calculations.

iii. Explain what the calculated Dunn index value indicates about the quality of the clusters.